

# IoT Might Disrupt Lots of Technologies for Greater Good

“I believe IoT will disrupt multiple industries in the coming decade and beyond,” says Manish Kothari, senior vice president of software development at Silicon Labs. In his keynote address at the VLSI Design Conference 2022, Manish talked about the challenges and future of IoT and its potential to be one of the most disruptive technologies of this century.



In today's time, safety and comfort are paramount, and automation is playing an active role in enhancing lifestyle, safety, and comfort along with improved efficiency. And to realise automation, gathering and processing data is the most crucial part, which is taken care of by the Internet of Things (IoT).

In a layman's terms, IoT is basically connecting different applications and sensors to the cloud or connecting them to the internet using a variety of different wireless protocols. The inclusion of IoT on a large scale has not just improved the quality of life but has also increased the complexity manifold.

## IoT is simple yet complex

IoT solutions seem straightforward but are not as simple as they sound. For example, an embedded system, which is the backbone of an IoT system, is often a low-powered device performing a simple task, which can easily be performed by even a low-cost smartphone, or a decade-old microcontroller (MCU) combined with a simple sensor that is easily available in the market.

The real challenge here is not to complete the task or gather data but to do all of it in a space-constrained, budget-constrained, power-constrained system that can

work for years without any human intervention. And not to forget about the security, which in an IoT system is paramount. Furthermore, there is a time-relevance aspect to the IoT solution, that is, the product and its components should remain relevant for at least 5 to 10 years.

“The IoT solution has at least a 5-to-10-year life,” says Manish. “So, there is a time relevance aspect of the hardware and software, both of which cannot go obsolete. You must be able to upgrade the capability of your devices, capabilities of your hardware and software, till the time an end customer has the product.”